

2022-Ph-H1-Q10

Section: Our Dynamic Universe

Topic: The Expanding Universe

Light from a distant galaxy is redshifted. What does this suggest?

Answer: E

(The galaxy is moving away from us)

Guidance for Students:

Redshift happens when the wavelength of light is stretched — a clear indicator the object emitting the light is **receding**. In cosmology, this is linked to the **expansion of the universe**.

Revision Tips:

- Redshift → object moving **away**
 - Blueshift → object moving **towards**
 - Hubble's Law: velocity $v = H_0 d$
 - Redshift gives evidence for the **Big Bang** and expanding space
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